

Numerical Analysis homework III

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I. Determine and check s is or not natural

We know $p(x)$ satisfies

$$\begin{cases} p(0) = 0 \\ p(1) = (2-x)^3|_{x=1} = 1 \\ p'(1) = ((2-x)^3)'|_{x=1} = -3(2-x)^2|_{x=1} = -3 \\ p''(1) = 6(2-x)|_{x=1} = 6 \end{cases}$$

We assume $p(x) = ax^3 + bx^2 + cx + d$, then

$$\begin{cases} d = 0 \\ a + b + c + d = 1 \\ 3a + 2b + c = -3 \\ 6a + 2b = 6 \end{cases} \Rightarrow \begin{cases} a = 7 \\ b = -18 \\ c = 12 \\ d = 0 \end{cases}$$

So $p(x) = 7x^3 - 18x^2 + 12x$.

$$s''(0) = -36 \neq 0$$

So $s(x)$ is NOT natural cubic spline.

II. Interpolation of f

II.a

Actually we have condition:

$$\begin{cases} p_i(x_i) = f_i & \& p_i(x_{i+1}) = f_{i+1}, i = 1, 2, \dots, n-1 \\ p'_i(x_{i+1}) = p'_{i+1}(x_{i+1}), i = 1, 2, \dots, n-2 \end{cases}$$

$3n - 4$ conditions in all. To determine s uniquely, by the theory of the rank of matrix we need $(n-1) * 3 = 3n - 3$ at least. So we need an additional condition to ensure the equation has the unique solution.

II.b

Assume $p_i(x) = a_i(x - x_i)^2 + b_i(x - x_i) + c_i$ by (a) we have

$$p_i(x_i) = f_i \quad \& \quad p_i(x_{i+1}) = f_{i+1} \quad \& \quad p'_i(x_i) = m_i$$

Then $\forall i = 1, 2, \dots, n-1$:

$$\begin{cases} p_i(x_i) = c_i = f_i \\ p'_i(x_i) = b_i = m_i \\ p_i(x_{i+1}) = a_i(x_{i+1} - x_i)^2 + b_i(x_{i+1} - x_i) + c_i = f_{i+1} \end{cases} \Rightarrow \begin{cases} a_i = \frac{f_{i+1} - f_i}{(x_{i+1} - x_i)^2} - \frac{m_i}{x_{i+1} - x_i} \\ b_i = m_i \\ c_i = f_i \end{cases}$$

Then $\forall i = 1, 2, \dots, n-1$, $p_i(x) = (\frac{f_{i+1} - f_i}{(x_{i+1} - x_i)^2} - \frac{m_i}{x_{i+1} - x_i})(x - x_i)^2 + m_i(x - x_i) + f_i$

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II.c

By (b) we get

$$m_{i+1} = p'_i(x_{i+1}) = 2 \frac{f_{i+1} - f_i}{x_{i+1} - x_i} - m_i$$

So we can compute m_2 , then $m_3, \dots, m_{n-1}$ by induction.

III. Determine a natural cubic spline

Assume $s_2(x) = ax^3 + bx^2 + dx + e$. Then we know

$$\begin{cases} s''_1(-1) = 0 \\ s''_2(1) = 0 \\ s_2(0) = s_1(0) = 1 + c \\ s'_2(0) = s'_1(0) = 1 + 3c \\ s''_2(0) = s''_1(0) = 6c \end{cases} \Rightarrow \begin{cases} 0 = 0 \\ 6a + 2b = 0 \\ e = 1 + c \\ d = 1 + 3c \\ 2b = 6c \end{cases} \Rightarrow \begin{cases} a = -c \\ e = 1 + c \\ d = 3c \\ b = 3c \end{cases}$$

So $s_2(x) = -cx^3 + 3cx^2 + 3cx + 1 + c$. Then if $s(1) = -1 \Rightarrow s_2(1) = -c + 3c + 3c + 1 + c = 6c + 1 = -1 \Rightarrow c = -\frac{1}{3}$.

IV. Cubic spline interpolation of given function

IV.a

We know

$$\begin{cases} f(-1) = 0 \\ f(0) = 1 \\ f(1) = 0 \end{cases}$$

Assume

$$s(x) = \begin{cases} p_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1, x \in [-1, 0] \\ p_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2, x \in [0, 1] \end{cases}$$

Then we get

$$\begin{cases} -a_1 + b_1 - c_1 + d_1 = 0 \\ d_1 = d_2 = 1 \\ a_2 + b_2 + c_2 + d_2 = 0 \\ c_1 = c_2 \\ b_1 = b_2 \\ 6a_2 + 2b_2 = 0 \\ -6a_1 + 2b_1 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = -\frac{1}{2} \\ a_2 = \frac{1}{2} \\ b_1 = -\frac{3}{2} \\ b_2 = -\frac{3}{2} \\ c_1 = 0 \\ c_2 = 0 \\ d_1 = 1 \\ d_2 = 1 \end{cases}$$

So

$$s(x) = \begin{cases} p_1(x) = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, x \in [-1, 0] \\ p_2(x) = \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, x \in [0, 1] \end{cases}$$

IV.b

By Newton's interpolation, we get $g(x) = 1 - x^2$. Then $g''(x) \equiv -2$. so

$$\int_{-1}^1 |g''(x)|^2 dx = 8 > \int_{-1}^1 |s''(x)|^2 dx = \int_0^1 |3x - 3|^2 dx + \int_{-1}^0 |3x + 3|^2 dx = 3 + 3 + 18 - 9 - 9 = 6$$

Then we verify that natural cubic splines have the minimal total bending energy.

V. B-spline

V.a

By the defination of B-spline

$$B_i^{n+1}(x) = \frac{x - t_{i-1}}{t_{i+n} - t_{i-1}} B_i^n(x) + \frac{t_{i+n+1} - x}{t_{i+n+1} - t_i} B_{i+1}^n(x)$$

$$B_i^0(x) = \begin{cases} 1, & x \in (t_{i-1}, t_i] \\ 0 & \text{otherwise} \end{cases}$$

By the defination of hat function

$$\hat{B}_i(x) = \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}}, & x \in (t_{i-1}, t_i] \\ \frac{t_{i+1}-x}{t_{i+1}-t_i}, & x \in (t_i, t_{i+1}] \\ 0 & \text{otherwise} \end{cases}$$

Then we know $B_i^1(x) = \hat{B}_i(x)$. Then we get

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} B_i^1(x) + \frac{t_{i+2} - x}{t_{i+2} - t_i} B_{i+1}^1(x) = \begin{cases} \frac{(x-t_{i-1})^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{(x-t_{i+2})^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0 & \text{otherwise} \end{cases}$$

V.b

Notice that

$$\frac{d}{dx} B_i^2(x) = \begin{cases} \frac{2(x-t_{i-1})}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} - \frac{(x-t_{i-1})}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} - \frac{(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{2(x-t_{i+2})}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0 & \text{otherwise} \end{cases}$$

We get

$$\begin{aligned} \frac{d}{dx} B_i^2(t_i^+) &= \frac{1}{(t_{i+1} - t_{i-1})} + \frac{1}{(t_{i+1} - t_i)} - \frac{(t_i - t_{i-1})}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} = \frac{2}{(t_{i+1} - t_{i-1})} \\ \frac{d}{dx} B_i^2(t_i^-) &= \frac{2}{(t_{i+1} - t_{i-1})} \\ \frac{d}{dx} B_i^2(t_{i+1}^-) &= \frac{(t_{i+2} - t_{i+1})}{(t_{i+2} - t_i)(t_{i+1} - t_i)} - \frac{1}{(t_{i+1} - t_i)} - \frac{1}{(t_{i+2} - t_i)} = -\frac{2}{(t_{i+2} - t_i)} \\ \frac{d}{dx} B_i^2(t_{i+1}^+) &= -\frac{2}{(t_{i+2} - t_i)} \end{aligned}$$

So $\frac{d}{dx} B_i^2(x)$ is continuous at t_i and t_{i+1} .

V.c

Notice that

$$\frac{d^2}{dx^2} B_i^2(x) = \begin{cases} \frac{2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{-2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{-2}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0 & \text{otherwise} \end{cases}$$

Besides, easy to find that $\frac{d}{dx} B_i^2(x) > 0, x \in (t_{i-1}, t_i]$ and $\frac{d^2}{dx^2} B_i^2(x) < 0, x \in (t_i, t_{i+1}]$, so suffices to show $\exists! x^* \in (t_i, t_{i+1})$, satisfied $\frac{d}{dx} B_i^2(x) = 0$. By (b), we get $\frac{d}{dx} B_i^2(t_i^+) > 0$ and $\frac{d}{dx} B_i^2(t_{i+1}^-) < 0$, so by intermediate value theorem of monotonic function, we know $\exists! x^* \in (t_i, t_{i+1})$, s.t. $\frac{d}{dx} B_i^2(x) = 0$.

$$\frac{(t_{i+1} - x)}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{(t_{i+2} - x)}{(t_{i+2} - t_i)(t_{i+1} - t_i)} - \frac{(x - t_{i-1})}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} - \frac{(x - t_i)}{(t_{i+2} - t_i)(t_{i+1} - t_i)} = 0$$

$\Leftrightarrow$

$$\frac{(t_{i+1} + t_{i-1} - 2x)}{(t_{i+1} - t_{i-1})} + \frac{(t_{i+2} + t_i - 2x)}{(t_{i+2} - t_i)} = 0$$

$\Leftrightarrow$

$$(t_{i+2} - t_i)(t_{i+1} + t_{i-1} - 2x) + (t_{i+1} - t_{i-1})(t_{i+2} + t_i - 2x) = 0$$

$\Leftrightarrow$

$$(t_{i+2} - t_i)(t_{i+1} + t_{i-1}) + (t_{i+1} - t_{i-1})(t_{i+2} + t_i) = 2x(t_{i+2} + t_{i+1} - t_{i-1} - t_i)$$

$\Rightarrow$

$$x^* = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{t_{i+2} + t_{i+1} - t_{i-1} - t_i}$$

V.d

$$B_i^2(x) = \begin{cases} 0 \leq \frac{(x-t_{i-1})^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})} < \frac{t_i-t_{i-1}}{t_{i+1}-t_{i-1}} < 1, x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, x \in (t_i, t_{i+1}] \\ 0 \leq \frac{(x-t_{i+2})^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})} < \frac{t_{i+2}-t_{i+1}}{t_{i+2}-t_i} < 1, x \in (t_{i+1}, t_{i+2}] \\ 0 \quad \text{otherwise} \end{cases}$$

suffices to show

$$\frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} \in [0, 1], x \in (t_i, t_{i+1}]$$

By (c) we know $B_i^2(x), x \in (t_i, x^*)$ monotonically increasing, $B_i^2(x), x \in (x^*, t_{i+1})$ monotonically decreasing.

$$0 \leq B_i^2(t_i) = \frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}} < 1$$

$$0 \leq B_i^2(t_i) = \frac{t_{i+2} - t_{i+1}}{t_{i+2} - t_i} < 1$$

So suffices to show $B_i^2(x^*) < 1$ i.e.

$$\frac{(x^* - t_{i-1})(t_{i+1} - x^*)}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{(x^* - t_i)(t_{i+2} - x^*)}{(t_{i+2} - t_i)(t_{i+1} - t_i)}$$

We know

$$x^* = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{t_{i+2} + t_{i+1} - t_{i-1} - t_i}.$$

Set

$$D := t_{i+2} + t_{i+1} - t_{i-1} - t_i.$$

$$\begin{aligned} x^* - t_{i-1} &= \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} - t_{i-1} = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i - t_{i-1}D}{D} = \frac{(t_{i+1} - t_{i-1})(t_{i+2} - t_{i-1})}{D}, \\ t_{i+1} - x^* &= t_{i+1} - \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} = \frac{t_{i+1}D - (t_{i+2}t_{i+1} - t_{i-1}t_i)}{D} = \frac{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)}{D}, \\ x^* - t_i &= \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} - t_i = \frac{t_{i+2}t_{i+1} - t_{i-1}t_i - t_iD}{D} = \frac{(t_{i+1} - t_i)(t_{i+2} - t_i)}{D}, \\ t_{i+2} - x^* &= t_{i+2} - \frac{t_{i+2}t_{i+1} - t_{i-1}t_i}{D} = \frac{t_{i+2}D - (t_{i+2}t_{i+1} - t_{i-1}t_i)}{D} = \frac{(t_{i+2} - t_{i-1})(t_{i+2} - t_i)}{D}. \end{aligned}$$

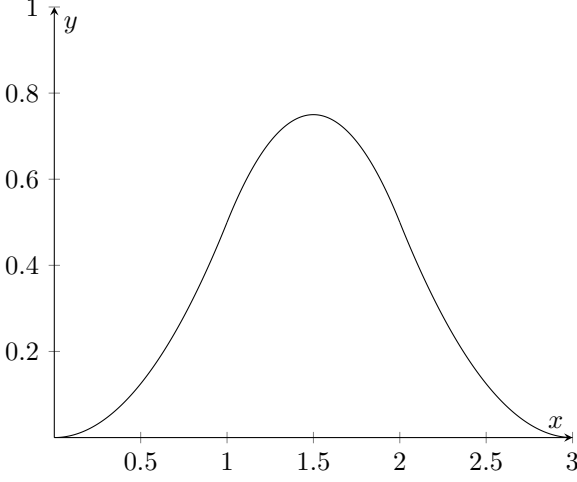
Then

$$\begin{aligned} B_i^2(x^*) &= \frac{\frac{(t_{i+1} - t_{i-1})(t_{i+2} - t_{i-1})}{D} \cdot \frac{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)}{D}}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{\frac{(t_{i+1} - t_i)(t_{i+2} - t_i)}{D} \cdot \frac{(t_{i+2} - t_{i-1})(t_{i+2} - t_i)}{D}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} \\ &= \frac{(t_{i+1} - t_{i-1})(t_{i+2} - t_{i-1})}{D^2} + \frac{(t_{i+2} - t_{i-1})(t_{i+2} - t_i)}{D^2} \\ &= \frac{t_{i+2} - t_{i-1}}{D^2} \left[(t_{i+1} - t_{i-1}) + (t_{i+2} - t_i) \right] = \frac{t_{i+2} - t_{i-1}}{D^2} (t_{i+1} + t_{i+2} - t_{i-1} - t_i) = \frac{t_{i+2} - t_{i-1}}{D}. \end{aligned}$$

So $B_i^2(x^*) = \frac{t_{i+2} - t_{i-1}}{t_{i+2} + t_{i+1} - t_{i-1} - t_i} < 1$.

V.e

We take $i = 1$ situation as an example. In general cases, only need to replace 0 with $i - 1$, 1 with i and so on.



VI. Verify a theorem

Firstly we know:

$$\begin{aligned}
 LHS &= [t_i, t_{i+1}, t_{i+2}](t-x)_+^2 - [t_{i-1}, t_i, t_{i+1}](t-x)_+^2 \\
 &= \frac{[t_{i+1}, t_{i+2}](t-x)_+^2 - [t_i, t_{i+1}](t-x)_+^2}{t_{i+2} - t_i} - \frac{[t_i, t_{i+1}](t-x)_+^2 - [t_{i-1}, t_i](t-x)_+^2}{t_{i+1} - t_{i-1}} \\
 &= \frac{(t_{i+2}-x)_+^2 - (t_{i+1}-x)_+^2}{(t_{i+2}-t_{i+1})(t_{i+2}-t_i)} - \frac{(t_{i+1}-x)_+^2 - (t_i-x)_+^2}{(t_{i+1}-t_i)(t_{i+2}-t_i)} - \frac{(t_{i+1}-x)_+^2 - (t_i-x)_+^2}{(t_{i+1}-t_i)(t_{i+1}-t_{i-1})} + \frac{(t_i-x)_+^2 - (t_{i-1}-x)_+^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})}
 \end{aligned}$$

When $x > t_{i+2}$ $LHS = 0 = RHS$.

When $x \leq t_{i-1}$

$$\begin{aligned}
 LHS &= \frac{(t_{i+2}-x)^2 - (t_{i+1}-x)^2}{(t_{i+2}-t_{i+1})(t_{i+2}-t_i)} - \frac{(t_{i+1}-x)^2 - (t_i-x)^2}{(t_{i+1}-t_i)(t_{i+2}-t_i)} - \frac{(t_{i+1}-x)^2 - (t_i-x)^2}{(t_{i+1}-t_i)(t_{i+1}-t_{i-1})} + \frac{(t_i-x)^2 - (t_{i-1}-x)^2}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})} \\
 &= \frac{t_{i+2} + t_{i+1} - 2x}{t_{i+2} - t_i} - \frac{t_{i+1} + t_i - 2x}{t_{i+2} - t_i} - \frac{t_{i+1} + t_i - 2x}{t_{i+1} - t_{i-1}} + \frac{t_i + t_{i-1} - 2x}{t_{i+1} - t_{i-1}} = 1 - 1 = 0
 \end{aligned}$$

When $t_{i-1} < x \leq t_i$

$$\begin{aligned}
 LHS &= \frac{t_{i+2} + t_{i+1} - 2x}{t_{i+2} - t_i} - \frac{t_{i+1} + t_i - 2x}{t_{i+2} - t_i} - \frac{t_{i+1} + t_i - 2x}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_i - t_{i-1})(t_{i+1} - t_{i-1})} \\
 &= \frac{2x - t_{i-1} - t_i}{t_{i+1} - t_{i-1}} + \frac{(t_i - x)^2}{(t_i - t_{i-1})(t_{i+1} - t_{i-1})} = \frac{(t_{i-1} - x)^2}{(t_i - t_{i-1})(t_{i+1} - t_{i-1})} = RHS
 \end{aligned}$$

When $t_i < x \leq t_{i+1}$

$$LHS = \frac{t_{i+2} + t_{i+1} - 2x}{t_{i+2} - t_i} - \frac{(t_{i+1} - x)^2}{(t_{i+1} - t_i)(t_{i+2} - t_i)} - \frac{(t_{i+1} - x)^2}{(t_{i+1} - t_i)(t_{i+1} - t_{i-1})}$$

at t_{i+1} $LHS|_{t_{i+1}} = \frac{t_{i+2}-t_{i+1}}{t_{i+2}-t_i} = RHS|_{t_{i+1}}$

Besides $\frac{d}{dx} LHS = \frac{2(t_{i+1}-x)}{(t_{i+1}-t_i)(t_{i+2}-t_i)} + \frac{2(t_{i+1}-x)}{(t_{i+1}-t_i)(t_{i+1}-t_{i-1})} - \frac{2}{t_{i+2}-t_i} = \frac{-2}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{-2}{(t_{i+2}-t_i)(t_{i+1}-t_i)} = \frac{d}{dx} RHS$

So $LHS = RHS, t_i < x \leq t_{i+1}$.

When $t_{i+1} < x \leq t_{i+2}$

$$LHS = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_{i+1})(t_{i+2} - t_i)} = RHS$$

Then $LHS = RHS$.

VII. Scaled integral of B-splines

By the derivatives of B-splines

$$\frac{d}{dx} B_i^{n+1}(x) = \frac{(n+1)B_i^n(x)}{t_{i+n} - t_{i-1}} - \frac{(n+1)B_{i+1}^n(x)}{t_{i+n+1} - t_i}$$

Since the support of $B_i^n(x)$ is finite $\Rightarrow$

$$\int_{\mathbb{R}} \frac{d}{dx} B_i^{n+1}(x) dx = \int_{\mathbb{R}} \frac{(n+1)B_i^n(x)}{t_{i+n} - t_{i-1}} dx - \int_{\mathbb{R}} \frac{(n+1)B_{i+1}^n(x)}{t_{i+n+1} - t_i} dx = 0$$

$\Rightarrow$

$$\frac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} B_i^n(x) dx = \frac{1}{t_{i+n+1} - t_i} \int_{t_i}^{t_{i+n+1}} B_{i+1}^n(x) dx$$

So the scaled integral of B-splines are independent of its index i .

VIII. Symmetric Polynomials

VIII.a